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Manuscript title: Meta-Prompted Instruction Refinement: Bridging Manual Prompting Techniques and Automatic Prompt Optimization

Manuscript type: Negative Result

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Dear Editor,

We submit our manuscript, “Meta-Prompted Instruction Refinement: Bridging Manual Prompting Techniques and Automatic Prompt Optimization,” for consideration by IEEE Access under the “Negative Result” manuscript type.

The paper reports a pre-registered go/no-go pilot testing Meta-Prompted Instruction Refinement (MPIR), a rubric-guided layer intended to refine automatic-prompt-optimization (APO) output for large language models. An initial evaluation using a closed, more capable model (GPT-4o) to judge and refine prompts for a weaker target model (GPT-3.5-turbo) reported per-task gains on a majority of Big-Bench Hard tasks, but a pooled effect that did not reach statistical significance — and a subsequent audit found its validation stage had scored candidates against non-held-out data. To determine whether the apparent gains reflected the refinement mechanism itself or the closed model’s capability advantage over its target, we reran the pilot with a single small open-weight model (Qwen3-1.7B) performing every role — target, optimizer, and refinement judge — on the three tasks where the original evaluation’s effect was largest. The effect did not reproduce: across all nine tested (optimizer, task) configurations, the refinement stage’s output was byte-identical to the unrefined baseline, and formal statistical tests (McNemar, GEE) returned no measurable effect in any of them.

We believe this fits IEEE Access and its “Negative Result” category specifically, for three reasons. First, the finding is a rigorously established boundary condition, not merely an unremarkable absence of effect: we trace the null result to a general defect in the refinement algorithm itself (it never scored the pre-refinement prompt as a baseline, so it could not have reported a null result even where one existed) and to two concrete failure modes in how a small model handles meta-prompting, each confirmed against live model output rather than assumed. Second, the result is directly actionable for practitioners: it identifies a specific, common deployment pattern (a single small open-weight model serving multiple roles) under which a popular class of cheap refinement layers should not be expected to help, and it identifies a narrower, lower-cost alternative (supplying a single worked exemplar to an otherwise exemplar-free optimizer) that our own pilot data suggests may close more of the accuracy gap than the refinement layer under test. Third, the paper documents seven real correctness defects found only by running the pipeline against a live model rather than a mocked one, each of general relevance to anyone building similar meta-prompting or refinement pipelines.

For full transparency: an earlier version of this work, evaluated under a different (closed-model) configuration later found to have a validation-data leak, was submitted to and desk-rejected by a different journal (*AI Open*) prior to this rebuild. The manuscript submitted here is a substantially revised and independently re-evaluated version reporting a different, corrected finding, not a resubmission of that earlier draft.

All code, configuration files, and per-example predictions underlying every number reported in this paper are publicly available at https://github.com/millennials92/meta_prompted_instruction_refinement, consistent with IEEE Access’s emphasis on replicability.

We confirm that this manuscript is not currently under review at any other venue, and that all authors have approved its submission to IEEE Access.

Thank you for your consideration.

On behalf of all authors,
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